

SIMATS ENGINEERING

CO3-ASSESSMENT TOOL1- CASE STUDY

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 4

Total Marks:40

Code of Conduct:

I (K.B.Devarshan/192511426) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- ii. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- iii. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

- (a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.
- (b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.
- (c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

- (a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.
- (b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.
- (c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$
- 2) P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$
- 3) P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts: $\text{SoilDry} = \text{True}, \text{CropWheat} = \text{True}$

- (a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).
- (b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.
- (c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) $\text{Stench}[1,2] \rightarrow \text{Wumpus is adjacent to cell}[1,2]$
- 2) $\text{Breeze}[1,1] \rightarrow \text{Pit is adjacent to cell}[1,1]$
- 3) $\text{Glitter}[x,y] \rightarrow \text{Gold is at}[x,y]$
- 4) $\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

QUESTIONS:

Case Study 1 — Smart Medical Diagnosis System

(a) **Propositional Logic Representation** We will assign propositional symbols to the medical conditions and facts:

- F: Patient has Fever
- C: Patient has Cough
- R: Patient has Rash

- B: Patient has Breathlessness
- Flu: Possible Flu
- M: Possible Measles
- P: Possible Pneumonia

Rules:

1. $F \wedge C \rightarrow \text{Flu}$
2. $F \wedge R \rightarrow M$
3. $C \wedge B \rightarrow P$

Facts for Patient A:

- $F = \text{True}$
- $C = \text{True}$
- $B = \text{True}$

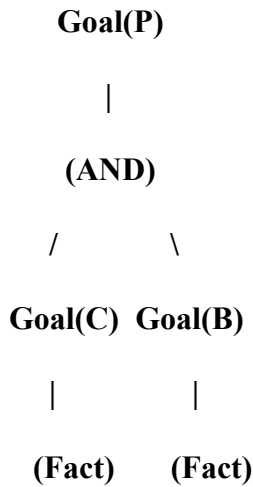
(b) Forward Chaining Forward chaining starts with the known facts and applies rules to infer new information until no more rules can be fired.

- **Initial Facts Base (FB):** $\{F, C, B\}$
- **Iteration 1:**
 - Evaluate Rule 1 ($F \wedge C \rightarrow \text{Flu}$): The antecedent $\{F, C\}$ is in the FB. The rule fires. Derived fact: Flu. New FB: $\{F, C, B, \text{Flu}\}$.
 - Evaluate Rule 2 ($F \wedge R \rightarrow M$): The antecedent requires R, which is not in the FB. Rule 2 does not fire.
 - Evaluate Rule 3 ($C \wedge B \rightarrow P$): The antecedent $\{C, B\}$ is in the FB. The rule fires. Derived fact: P. New FB: $\{F, C, B, \text{Flu}, P\}$.
- **Iteration 2:**
 - No new rules can be fired because no new antecedents are fully satisfied.
- **Conclusion:** The possible diagnoses for Patient A are **Possible Flu** (Flu) and **Possible Pneumonia** (P).

(c) Backward Chaining Goal Reduction Tree The goal is to prove **Patient A has Pneumonia (P)**. Backward chaining works from the goal back to the initial facts.

- **Goal:** P
- Rule 3 concludes P: $C \wedge B \rightarrow P$.
- To prove P, we must prove the sub-goals C and B.
- Is C known? Yes, it is a given fact.
- Is B known? Yes, it is a given fact.

- **Goal Reduction Tree:**



Case Study 2 — Autonomous Traffic Management Agent

(a) Unification Rule 1 is: For all x (Vehicle(x) AND EmergencyType(x) implies ClearPath(x)).

- Facts given: Vehicle(Ambulance), EmergencyType(Ambulance).
- Step 1: Match Vehicle(x) with Vehicle(Ambulance).
 - Substitution: $\Theta_1 = \{x = \text{Ambulance}\}$
- Step 2: Apply Θ_1 to the rest of the antecedent: EmergencyType(Ambulance).
- Step 3: Match derived EmergencyType(Ambulance) with the fact EmergencyType(Ambulance). It is a perfect match.
- Complete Substitution Set (MGU): $\Theta = \{x = \text{Ambulance}\}$

(b) Resolution Method

First, convert the KB and the negated goal into Conjunctive Normal Form (CNF).

Rules in CNF:

- R1: NOT Vehicle(x) OR NOT EmergencyType(x) OR ClearPath(x)
- R2a: NOT ClearPath(x) OR GreenSignal(x)
- R2b: NOT ClearPath(x) OR NOT RedSignal(x)
- R3: NOT Vehicle(x) OR NOT Behind(x,y) OR NOT GreenSignal(y) OR Proceed(x)

Facts in CNF:

- F1: Vehicle(Ambulance)
- F2: EmergencyType(Ambulance)
- F3: Vehicle(CarA)
- F4: Behind(CarA, Ambulance)

Negated Goal (NG): NOT Proceed(CarA)

Resolution Steps:

1. Resolve R3 with NG using $\{x = \text{CarA}\}$: Yields NOT Vehicle(CarA) OR NOT Behind(CarA,y) OR NOT GreenSignal(y)
 2. Resolve with F3: Yields NOT Behind(CarA,y) OR NOT GreenSignal(y)
 3. Resolve with F4 using $\{y = \text{Ambulance}\}$: Yields NOT GreenSignal(Ambulance)
 4. Resolve R1 with F1 using $\{x = \text{Ambulance}\}$: Yields NOT EmergencyType(Ambulance) OR ClearPath(Ambulance)
 5. Resolve with F2: Yields ClearPath(Ambulance)
 6. Resolve R2a with Step 5: Yields GreenSignal(Ambulance)
 7. Resolve Step 6 with Step 3: Yields the Empty Clause.
- Conclusion: Since deriving the empty clause creates a contradiction, Proceed(CarA) is proven true.

(c) Analysis of FOL KB for Simultaneous Emergencies The current FOL knowledge base is insufficient to handle two simultaneous emergency vehicles.

- Justification: The rule stating that a ClearPath gives a GreenSignal applies to any emergency vehicle. If two emergency vehicles approach the same intersection from perpendicular directions, the system would attempt to give conflicting green signals simultaneously, leading to a collision.
- Additional Rules Needed: The KB lacks spatial and conflict-resolution context. It requires rules to determine intersection location (e.g., Approaching(x, Intersection)) and priority rules (e.g., HigherPriority(x, y)) to ensure that both vehicles do not get a green signal at the same crossing at the same time.

Case Study 3 — Agricultural Expert Reasoning System

(a) Conversion to Conjunctive Normal Form (CNF) Let: S = SoilDry, I = IrrigationNeeded, W = CropWheat, A = ApplyDripMethod, R = CropAtRisk.

- P1: S implies I
 - Eliminate implication: NOT S OR I
- P2: I AND W implies A
 - Eliminate implication: NOT (I AND W) OR A
 - Move NOT inward (De Morgan's): NOT I OR NOT W OR A
- P3: NOT A AND W implies R
 - Eliminate implication: NOT (NOT A AND W) OR R
 - Move NOT inward: A OR NOT W OR R
- Facts: S, W

(b) Resolution Refutation for 'ApplyDripMethod' (A)

- Negated Goal: NOT A
- Clauses:
 - NOT S OR I
 - NOT I OR NOT W OR A
 - A OR NOT W OR R

- S
- W
- NOT A (Negated Goal)
- Resolution Steps:
 - Resolve Clause 1 (NOT S OR I) with Clause 4 (S) to derive I.
 - Resolve Clause 2 (NOT I OR NOT W OR A) with derived I to derive NOT W OR A.
 - Resolve derived NOT W OR A with Clause 5 (W) to derive A.
 - Resolve derived A with Clause 6 (NOT A) to derive the Empty Clause.
- Because we reached the empty clause, the theorem ApplyDripMethod (A) is proven true.

(c) Removing the Fact 'SoilDry'

- If the fact SoilDry (S) is removed from the KB, we can no longer deduce IrrigationNeeded (I) via P1.
- Without I, P2 (NOT I OR NOT W OR A) cannot be satisfied to derive A.
- Does 'CropAtRisk' (R) still follow? No. To logically derive R using P3 (A OR NOT W OR R), the system must know that NOT A is unequivocally true. In standard logic, the inability to prove A does not automatically mean NOT A is true. Since NOT A is not an explicitly stated fact or inferable rule, CropAtRisk cannot be logically concluded.

Case Study 4 — Wumpus World Knowledge Agent

(a) Constructing the Agent's Belief State Given the rules in the KB, we apply Modus Ponens to the percepts:

1. Percept: Stench at [1,2]
 - Rule: Stench at [1,2] implies Wumpus is adjacent to cell [1,2]
 - Conclusion: The Wumpus is adjacent to cell [1,2].
2. Percept: Breeze at [1,1]
 - Rule: Breeze at [1,1] implies Pit is adjacent to cell [1,1]
 - Conclusion: A Pit is adjacent to cell [1,1].
3. Percept: Glitter at [2,2]
 - Rule: Glitter at [2,2] implies Gold is at [2,2]
 - Conclusion: Gold is at cell [2,2].

(b) Forward Chaining for Wumpus and Pit Locations

Based on Wumpus World topology (where adjacent cells share an edge):

- Locating the Pit: The Pit is adjacent to [1,1]. Adjacent cells to [1,1] are [1,2] and [2,1].
 - Since the agent reached [1,2] to perceive the Stench (and did not die), [1,2] cannot contain a Pit.
 - Therefore, the Pit must be located at [2,1].
- Locating the Wumpus: The Wumpus is adjacent to [1,2]. Adjacent cells to [1,2] are [1,1], [1,3], and [2,2].
 - [1,1] is safe (the agent survived there).
 - [2,2] is where the agent perceives Glitter (implying the agent is there and safe).
 - Therefore, the Wumpus must be located at [1,3].

(c) Evaluating Path Safety: [1,1] to [1,2] to [2,2] The path is safe based on the rule NOT Wumpus at $[x,y]$ AND NOT Pit at $[x,y]$ implies Safe at $[x,y]$.

- Cell [1,1]: It is the starting point. By definition, if the agent acts from here, NOT Wumpus at [1,1] AND NOT Pit at [1,1] is true. The cell is Safe.
- Cell [1,2]: The agent deduced the Pit is at [2,1] and the Wumpus is at [1,3]. Therefore, NOT Wumpus at [1,2] AND NOT Pit at [1,2] is true. The cell is Safe.
- Cell [2,2]: The agent deduced the Pit is at [2,1] and the Wumpus is at [1,3]. Furthermore, the agent safely perceives Gold (Glitter) at this specific cell. Therefore, NOT Wumpus at [2,2] AND NOT Pit at [2,2] is true. The cell is Safe.
- Conclusion: Logical inference confirms that every node along the path lacks both a Pit and a Wumpus, making it fully safe for the agent to navigate and collect the gold.